Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment

A Report submitted in partial fulfillment of the requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering (Internet of Things)

by

Jhansi Singavarapu 2111CS050024
Madhu Maheshuni 2111CS050031
Aravindh Murkham 2111CS050012

Under the esteemed guidance of

Mrs. N. Shalini

Assistant Professor



Department of Computer Science and Engineering (Internet of Things)

School of Engineering

MALLA REDDY UNIVERSITY

Maisammaguda, Dulapally, Hyderabad, Telangana 500100

2025

Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment

A Report submitted in partial fulfillment of the requirements for the Degree of

Bachelor of Technology

in

Computer Science and Engineering (Internet of Things)

by

Jhansi Singavarapu 2111CS050024

Madhu Maheshuni 2111CS050031

Aravindh Murkham 2111CS050012

Under the esteemed guidance of

Mrs. N. Shalini

Assistant Professor



Department of Computer Science and Engineering (Internet of Things)

School of Engineering

MALLA REDDY UNIVERSITY

Maisammaguda, Dulapally, Hyderabad, Telangana 500100

2025

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

Department of Computer Science and Engineering (Internet of Things)

CERTIFICATE

This is to certify that the project report entitled **"Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment"**, submitted by **Singavarapu Jhansi(2111CS050024), Madhu Maheshuni (2111CS050031), Aravindh Murkham (2111CS050012),** towards the partial fulfillment for the award of Bachelor's Degree in Computer Science and Engineering – Internet of Things from the Department of CS & IOT

, Malla Reddy University, Hyderabad, is a record of bonafide work done by him/ her. The results embodied in the work are not submitted to any other University or Institute for award of any degree or diploma.

Internal Guide
Mrs. N. Shalini
Assistant Professor

Head of the Department
Dr. G. Anand Kumar
CSE(Cyber Security & IoT)

External Examiner

DECLARATION

We hereby declare that the project report entitled **"Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment"** has been carried out by us and this work has been submitted to the **Department of Computer Science and Engineering (Internet of Things), Malla Reddy University,** Hyderabad in partial fulfillment of the requirements for the award of degree of Bachelor of Technology. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

Place:

Date:

Jhansi Singavarapu 2111CS050024

Madhu Maheshuni 2111CS050031

Aravindh Murkham 2111CS050012

ACKNOWLEDGEMENT

We extend our sincere gratitude to all those who have contributed to the completion of this project report. Firstly, We would like to extend our gratitude to **Dr. V. S. K Reddy, Vice-Chancellor**, for his visionary leadership and unwavering commitment to academic excellence.

We would also like to express my deepest appreciation to our project **Mrs. N. Shalini, Assistant Professor**, whose invaluable guidance, insightful feedback, and unwavering support have been instrumental throughout the course of this project for successful outcomes.

We extend our gratitude to **Dr. G. Latha, PRC-convenor,** for giving valuable inputs and timely guidelines to improve the quality of our project through a critical review process. We thank our project coordinator **Dr. Nageshwar Rao,** for his timely support.

We are also grateful to **Dr. G. Anand Kumar, Head of the Department of Internet of Things,** for providing us with the necessary resources and facilities to carry out this project .

We are deeply indebted to all of them for their support, encouragement, and guidance, without which this project would not have been possible.

ABSTRACT

The Smart Car Seat Belt Safety System is designed to enhance passenger safety by monitoring vital signs and detecting accidents in real time. This innovative system utilizes an Arduino Uno microcontroller integrated with a temperature and humidity sensor (DHT11), pulse sensor, position sensor, and vibration sensor. The system continuously monitors the occupant's health and environmental conditions. In the event of an accident or abnormal health conditions, the vibration sensor detects impact, and the pulse sensor tracks the occupant's heart rate. The GPS module records the exact location, while the GSM module sends an SMS alert to pre-defined contacts, providing real-time information about the accident and location. Additionally, the collected data is sent to the ThingSpeak cloud for remote monitoring and analysis. This system ensures timely response during emergencies, thereby reducing fatality risks. The Smart Car Seat Belt Safety System demonstrates a reliable and efficient approach to intelligent vehicle safety, emphasizing passenger well-being and accident management.

INDEX

Contents	Page No.
Chapter 1 Introduction	
1.1 Problem Definition & Description	1
1.2 Objective of the project	2
1.3 Scope of the Project	3
Chapter 2 System Analysis	4-15
2.1 Existing System	4
2.1.1 Background & Literature Survey	5-6
2.1.2 Limitations of Existing System	6-7
2.2 Proposed System	7
2.2.1 Advantages of Proposed System	7
2.3 Software & Hardware Requirements	8
2.3.1 Software Requirements	8
2.3.2 Hardware Requirements	9-14
2.4 Feasibility Study	14-15
2.4.1 Technical Feasibility	14
2.4.2 Economic Feasibility	15
2.4.3 Legal and Ethical Feasibility	15
2.4.4 Operational Feasibility	15
2.4.5 SocialFeasibility	15
Chapter 3 Architectural Design	16-18
3.1 Block Diagram	16
3.1.1 Methodology	16-17
3.2 Data flow & Process flow Diagram	17-18
Chapter 4 Implementation	19-29

4.1 Coding Blocks 19-29
Chapter 5 Testing & Results 30-34
5.1 Prototype 30-32
5.2 Resulting Graphs 33-34
Chapter 6 Conclusions & Future Scope 35
6.1 Conclusion 35
6.1.1 Future Scope 35
Bibliography 36-37
Paper Publication 38-41

Github Repository : *https://github.com/singajhansi/-Smart-Car-Seat-Belt-Accident-Detection-and-Emergency-Services-in-Smart-City-Environment*

LIST OF FIGURES

Contents		Page No
Figure 1	Arduino UNO	10
Figure 2	16x2 LCD display	10
Figure 3	Wifi	11
Figure 4	MEMES sensor	12
Figure 5	DHT 11 Sensor	12
Figure 6	Pulse Sensor	13
Figure 7	Vibration Sensor	13
Figure 8	GSm Module	14
Figure 9	Block Diagram	16
Figure 10	Methodology	17
Figure 11	Data flow Diagram	18
Figure 12	Prototype	30
Figure 13	Power supply to prototype	31
Figure 14	LCD Power ON	31
Figure 15	Pulse rate Checking	31
Figure 16	DHT Checking	32
Figure 17	SMS sending	32
Figure 18	Alert Received	32
Figure 19	Temperature Graph	33
Figure 20	Humidity Graph	33
Figure 21	Pulse Graph	34
Figure 22	Vibration Sensor	34
Figure 23	Date and Time Graph	34

CHAPTER - 1 INTRODUCTION

1.1 PROBLEM DEFINITION & DESCRIPTION

Road accidents are a major cause of fatalities worldwide, and ensuring passenger safety is a top priority in modern vehicle systems. Traditional seat belts, while effective in minimizing injuries during collisions, lack the capability to monitor the occupant's health and environmental conditions. In today's rapidly advancing technological era, smart systems have become essential in promoting safety and real-time monitoring. The Smart Car Seat Belt Safety System leverages cutting edge embedded technology to address this challenge by integrating health monitoring, accident detection, and real-time alert mechanisms.

The primary objective of this system is to monitor the passenger's vital signs, detect accidents, and promptly transmit crucial data to emergency contacts and cloud platforms. This comprehensive safety solution is built around an Arduino Uno microcontroller, which acts as the core processing unit for data acquisition and communication. The system incorporates multiple sensors, including a temperature and humidity sensor (DHT11) to monitor environmental conditions, a pulse sensor to track the occupant's heart rate, and a position sensor to detect seat belt engagement. Additionally, a vibration sensor is employed to detect collisions and sudden impacts that may indicate an accident.

In the event of a collision or any abnormal health condition, the system promptly activates the emergency alert mechanism. The GPS module accurately records the vehicle's location, while the GSM module sends an SMS alert containing location coordinates and vital information to pre-defined emergency contacts, including family members and medical assistance teams. This timely response can significantly reduce the risk of fatalities by ensuring that help arrives as quickly as possible.

Furthermore, the system utilizes the ThingSpeak cloud platform to transmit real time data for remote monitoring and data analysis. By uploading health parameters, environmental conditions, and accident alerts to the cloud, authorized personnel can continuously monitor the occupant's safety and analyze data for long-term safety improvements. This feature proves particularly useful for fleet management, public transport systems, and personal vehicle monitoring.

The Smart Car Seat Belt Safety System is designed to be compact, cost-effective, and easily deployable in various vehicles. By integrating health monitoring, accident detection, and real-time data transmission, this system addresses the growing need for intelligent safety solutions in the automotive sector. The use of Arduino Uno enhances flexibility and modularity, making it feasible to upgrade and customize the system based on specific safety requirements.

In conclusion, the Smart Car Seat Belt Safety System represents a significant leap towards intelligent transportation safety, combining health monitoring with accident detection and instant communication. This innovation promises to improve the safety standards of modern vehicles while safeguarding the lives of occupants through real-time intervention and monitoring.

1.2 OBJECTIVES OF THE PROJECT

The primary objective of the Smart Car Seat Belt Safety System is to enhance passenger safety by continuously monitoring vital signs and detecting accidents in real-time. This system aims to integrate health monitoring, environmental sensing, accident detection, and real-time communication using Arduino Uno, GPS, GSM, and cloud connectivity. The system ensures timely intervention by sending SMS alerts with GPS coordinates to emergency contacts while uploading critical data to the ThingSpeak cloud for remote monitoring and analysis. This innovative approach aims to minimize fatality risks and improve vehicle safety standards effectively.

1.3 SCOPE OF THE PROJECT

The project "Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment" aims to develop an intelligent system that enhances passenger safety by detecting accidents in real time and automatically alerting emergency services. The system comprises a smart car seat belt equipped with embedded sensors such as pressure sensors, accelerometers, and impact sensors, integrated with a microcontroller or processor like Arduino or Raspberry Pi. In case of a collision, the system promptly transmits accident data, including location and severity, to emergency services using communication modules like GSM, GPS, or IoT-based technologies. It also notifies car owners or family members. The system leverages IoT protocols, cloud platforms, and machine learning algorithms to analyze accident patterns and enhance prediction accuracy. By integrating with smart city infrastructure, the system facilitates efficient data management and rapid response, contributing to safer and smarter urban environments. The implementation follows a systematic approach, including requirement analysis, hardware integration, software development, and testing, culminating in deployment and performance evaluation. Through this project, the response time for emergency services is significantly reduced, improving passenger safety and public trust in smart city initiatives.

CHAPTER - 2 SYSTEM ANALYSIS

System Analysis is the process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components. It is conducted for the purpose of studying a system or its parts in order to identify its objectives. It is a problem solving technique that improves the system and ensures that all the components of the system work efficiently to accomplish their purpose. Analysis specifies what the system should do.

2.1 EXISTING SYSTEM

Existing systems for accident detection and emergency response in vehicles mainly consist of *Automatic Crash Notification (ACN)* systems and basic airbag deployment alerts. These systems use sensors like accelerometers and gyroscopes to detect sudden impacts and trigger emergency responses, such as deploying airbags and sending crash notifications to designated contacts or emergency services. Some advanced vehicles are equipped with *eCall* systems (especially in Europe), which automatically dial emergency numbers and transmit location data. However, these systems have several limitations. They typically focus on impact detection without actively monitoring seat belt usage or considering pre-impact data, which could help assess crash severity more accurately. Moreover, they often rely on cellular connectivity without leveraging modern IoT protocols, which can result in communication delays or failures, especially in areas with poor network coverage.

Another significant drawback is the lack of integration with smart city infrastructure, which could enable faster coordination between emergency services and hospitals. Additionally, existing systems are mostly limited to highend vehicles, making them inaccessible to a broader population. As a result, response times may still be delayed, and crucial information regarding the

accident may not be effectively communicated to rescue teams. The proposed system aims to address these gaps by integrating real-time IoT-based monitoring, automated emergency alerts, and smart city connectivity to significantly enhance safety and response efficiency.

2.1.1 BACKGROUND AND LITERATURE SURVEY

1. TITLE: Smart Vehicle Accident Prevention and Road Safety System with Real Time Data Acquisition https://www.mecs-press.org/ijem/ijem-v14-n3/v14n3-2.html

ABSTRACT: Smart Vehicle Accident Prevention System is an innovative solution aimed at enhancing road safety and reducing the occurrence of accidents. Leveraging the Internet of Things (IoT) technology, this system combines realtime data acquisition, analysis, and intelligent decision-making algorithms to provide an effective accident prevention mechanism. The Vehicle Accident Prevention System is a com-prehensive project that aims to enhance road safety by utilizing Arduino microcontrollers and various sensors, including an alcohol sensor, temperature sensor, IR sensor and ultrasonic sensor. This report provides a detailed overview of the system's design, implementation, and functionality.

2. TITLE: IoT Based Design of Automatic Seat Belt System for Vehicles https://www.researchgate.net/publication/354391411_IoT_Based_Design_of_Automatic_Seat_Belt_System_for_Vehicles

ABSTRACT: Seats belts are one of the important components in the passenger's safety. It is designed in such a way that to avoid the travelers from being thrown out of the car. It is specially designed to analyze the crash impact and it avoids the collision between each other during any accidents. Three - point seat belt is the most common and mandatory safety system in street legal automobile, developed by Volvo engineer Nils Bohlin in 1959.we realize that, although it saves a lot of lives in accidents, in certain unlikely event it can take life too.

This can happen as the seat belt may get jammed causing the passengers to be stuck on their seats. This may lead to a dangerous situation where the driver and the passengers to be in grave danger. Our objective is to design an improvised Three-point seat belt unclamping mechanism, which includes controlled release of seat belt buckle. This is controlled by on board ECU (electronic control unit). The project involves usage of 3D modelling software Autodesk inventor.

3. TITLE: VEHICLE ACCIDENT DETECTION SYSTEM BY USING GSM AND GPS

https://www.irjet.net/archives/V6/i1/IRJET-V6I1294.pdf

ABSTRACT: -Arduino Based Vehicle Accident Alert System using GPS, GSM and Accelerometer. Accelerometer detects the sudden change in the axes of vehicle and GSM module send the alert message on your Mobile Phone with the location of the accident. The advancing technology has made our day to day lives easier. Since every coin has two sides similarly technology has its benefits as well as its disadvantages. The rise in technology has increased the rate of road accidents which causes huge loss of life. The poor emergency facilities available in our country just add to this problem. Our project is going to provide a solution to this problem.

2.1.2 LIMITATIONS OF EXISTING SYSTEM

1. **Delayed Emergency Response:** Traditional accident detection systems may take longer to notify emergency services, leading to delays in providing medical assistance.

2. **Manual Reporting Dependency**: Many systems still rely on manual reporting by witnesses or passengers, which may not be possible if the occupants are unconscious or incapacitated.

3. **Lack of Real-Time Monitoring:** Conventional seat belt systems do not continuously monitor the driver and passenger's health status or the car's realtime conditions, limiting proactive accident prevention.

4. **Inaccurate Impact Detection:** Some systems might generate false positives or fail to detect certain types of collisions, especially minor but potentially harmful impacts.

5. **Limited Integration with Smart City Infrastructure:** Existing solutions often lack seamless integration with smart city ecosystems, such as real-time traffic monitoring or automated ambulance dispatch.

6. **No Driver Condition Monitoring:** Current systems do not assess whether the driver is wearing a seat belt or detect driver fatigue, which are crucial for accident prevention.

2.2 PROPOSED SYSTEM

The proposed Smart Car Seat Belt Safety System integrates multiple sensors with an Arduino Uno microcontroller to monitor the occupant's vital signs, environmental conditions, and detect accidents in real time. The system uses a temperature and humidity sensor (DHT11), pulse sensor, position sensor, and vibration sensor to gather data. In the event of an accident or abnormal health condition, the vibration sensor detects impact, while the GPS module records the location. The GSM module then sends an SMS alert containing location data to emergency contacts. Additionally, the collected data is transmitted to the ThingSpeak cloud for real-time monitoring and analysis. This system is designed to ensure rapid response during emergencies, thereby enhancing passenger safety and reducing fatality risks.

2.2.1 ADVANTAGES OF PROPOSED SYSTEM

- Enhanced Passenger Safety: Monitors vital signs and environmental conditions continuously to ensure occupant well-being.
- Real-Time Accident Detection: Vibration sensor detects collisions and instantly triggers emergency alert mechanisms.
- Immediate Emergency Response: Sends SMS alerts with GPS coordinates to emergency contacts for quick assistance.
- Cloud Data Monitoring: Uploads real-time data to the ThingSpeak platform for remote safety monitoring and analysis.
- Cost-Effective Solution: Uses Arduino-based components, making it affordable and easily deployable in various vehicles.

2.3 HARDWARE AND SOFTWARE REQUIREMENTS

Every computer needs software to function, a computer must have specific hardware or additional software resources installed. These prerequisites are referred to as (computer) system requirements, and they are frequently utilised as recommendations rather than strict guidelines. The majority of software specifies minimum and recommended system requirements. System requirements typically rise with time due to the growing need for more processing power and resources in newer software versions. Industry observers say that rather than technological breakthroughs, this trend is a more significant factor in the improvements of current computer systems.

2.3.1 SOFTWARE REQUIREMENTS

A software requirements specification (SRS) is a comprehensive description of the intended purpose and environment for software under development.

- Arduino ide
- Embedded c programming
- Windows OS

2.3.2 HARDWARE REQUIREMENTS

These requirements include the minimum processor speed, memory, and disk space required for the implementation of this project.

- ARDUINO UNO
- MEMES SENSOR
- DHT11
- PULSE SENSOR
- VIBRATION SENSOR
- LCD
- WIFI
- GSM
- REGULATED POWER SUPPLY

DESCRIPTION OF TOOLS

1. ARDUINO UNO

The most common version of Arduino is the Arduino Uno. This board is what most people are talking about when they refer to an Arduino. The Uno is one of the more popular boards in the Arduino family and a great choice for beginners. There are different revisions of Arduino Uno, below detail is the most recent revision (Rev3 or R3).

The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

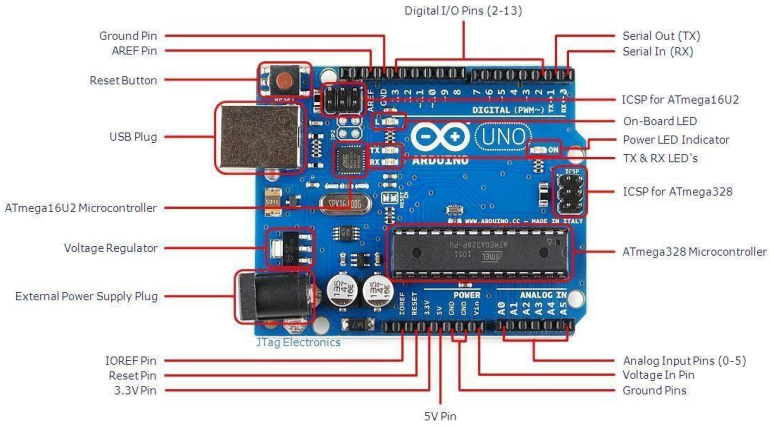


Fig 1 : Arduino UNO

2. LCD

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animationsand so on.

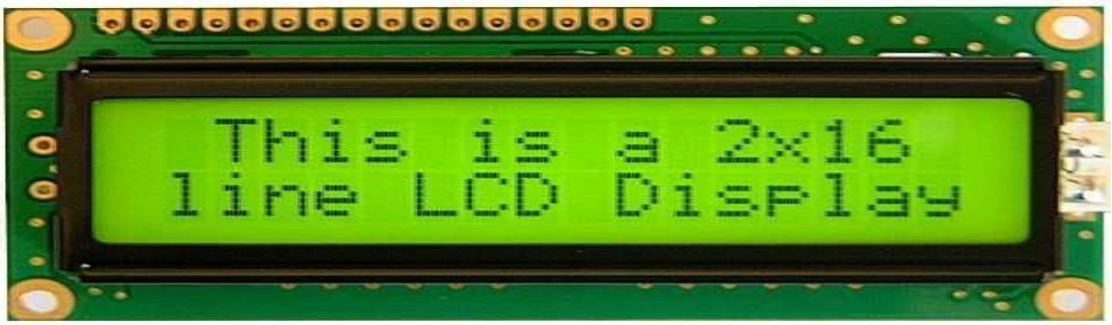


Fig 2 : 16x2 LCD display

3. WIFI

The **ESP8266** is a low-cost Wi-Fi chip with full TCP/IP stack and MCU (microcontroller unit) capability produced by Shanghai-based Chinese manufacturer, Espressif Systems.[1]

The chip first came to the attention of western makers in August 2014 with the ESP-01 module, made by a third-party manufacturer, AI-Thinker. This small module allows microcontrollers to connect to a Wi-Fi network and make simple TCP/IP connections using Hayes-style commands. However, at the time there was almost no English-language documentation on the chip and the commands it accepted.[2] The very low price and the fact that there were very few external components on the module which suggested that it could eventually be very inexpensive in volume, attracted many hackers to explore the module, chip, and the software on it, as well as to translate the Chinese documentation.[3]



Fig 3 : WIFI

4. MEMES SENSOR



Fig 4 : MEMES Sensor

The **MEMS sensor** in the Smart Car Seat Belt Safety System acts as an **accelerometer** to detect sudden movements or impacts, indicating potential accidents. It measures changes in velocity or orientation by detecting linear acceleration or gravitational forces. The **Arduino Uno** processes the sensor's electrical signals to detect collisions and triggers the **GSM module** to send emergency SMS alerts. Additionally, MEMS sensors can monitor seat belt positioning to ensure it is fastened correctly.

5. DHT11:

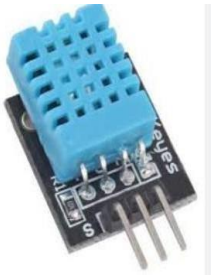


Fig 5 : DHT11 Sensor

The **DHT11 sensor** in the Smart Car Seat Belt Safety System monitors temperature and humidity to ensure passenger comfort and safety. It uses a **thermistor** for temperature and a **capacitive humidity sensor** for moisture detection, communicating with the **Arduino Uno** via a single-wire digital interface. With a response time of about 2 seconds, it provides real-time

monitoring. If abnormal conditions are detected, the system can alert the driver or take automated actions, and data can be sent to the **ThingSpeak cloud** for remote monitoring and analysis.

6. PULSE SENSOR



Fig 6 : Pulse Sensor

The **Pulse Sensor** in the Smart Car Seat Belt Safety System monitors the passenger's heart rate to detect health abnormalities. It emits infrared light through the skin and measures the reflected light to calculate the pulse rate. Connected to the **Arduino Uno**, it continuously tracks heart rate and triggers alerts if dangerous levels are detected, including sounding a buzzer and sending emergency SMS via the **GSM module**. This real-time health monitoring enhances safety and helps prevent medical emergencies on the road.

7. VIBRATION SENSOR



Fig 7 : Vibration Sensor

The **Vibration Sensor** in the Smart Car Seat Belt Safety System detects collisions or sudden impacts, triggering the emergency response mechanism. Typically, a **piezoelectric vibration sensor** is used to generate an electric signal proportional to vibration intensity. The **Arduino Uno** processes this signal to determine if it exceeds a threshold. Upon impact detection, the system activates the **GPS**

module to record the location and the **GSM module** to send an SMS alert to emergency contacts.

8. GSM



Fig 8 : GSM module

The **GSM module** in the Smart Car Seat Belt Safety System enables communication by sending emergency SMS alerts when an accident or health abnormality is detected. Common modules like **SIM800L** or **SIM900** interface with Arduino via **UART** communication, using **AT commands** for operations like sending SMS and making calls. During a collision or health emergency, the system gathers data from sensors (e.g., accelerometer, pulse sensor) and sends GPS coordinates and vital information to emergency contacts.

2.4 FEASIBILITY STUDY

2.4.1 Technical Feasibility:

- **Advanced Sensor Integration:** Utilizing accelerometers, gyroscopes, pressure sensors, and GPS to detect accidents and seat belt usage.
- **Data Communication:** Employing IoT protocols (like MQTT or CoAP) to transmit accident data to emergency services.
- **Cloud and Edge Computing:** Ensuring quick data processing and decision-making through edge devices and cloud platforms.
- **Smart City Integration:** Connecting with traffic management systems, healthcare facilities, and law enforcement for a coordinated response.

2.4.2 Economic Feasibility:

- **Initial Costs:** Expenses related to sensor installation, connectivity modules, and integration with city infrastructure.
- **Operational Costs:** Maintenance, data storage, and real-time monitoring.
- **Cost-Benefit Analysis:** Reduced fatalities and faster emergency response outweigh initial and operational costs.

2.4.3. Legal and Ethical Feasibility:

- **Data Privacy:** Ensuring compliance with data protection laws (e.g., GDPR) to safeguard personal information.
- **Liability Concerns:** Defining accountability in case of system failure or false positives.
- **Ethical Issues:** Avoiding misuse of accident data for surveillance or tracking.

2.4.4. Operational Feasibility:

- **Ease of Use:** The system must be automated to reduce driver involvement during accidents.
- **Maintenance and Reliability:** Regular updates and maintenance of hardware and software are necessary.
- **Training and Awareness:** Educating users and emergency personnel about the system's functionality.

2.4.5. Social Feasibility:

- **Public Acceptance:** Gaining public trust by ensuring privacy and safety.
- **Awareness Campaigns:** Promoting the benefits to encourage widespread adoption.

CHAPTER - 3 ARCHITECTURAL DESIGN

3.1 Block Diagram

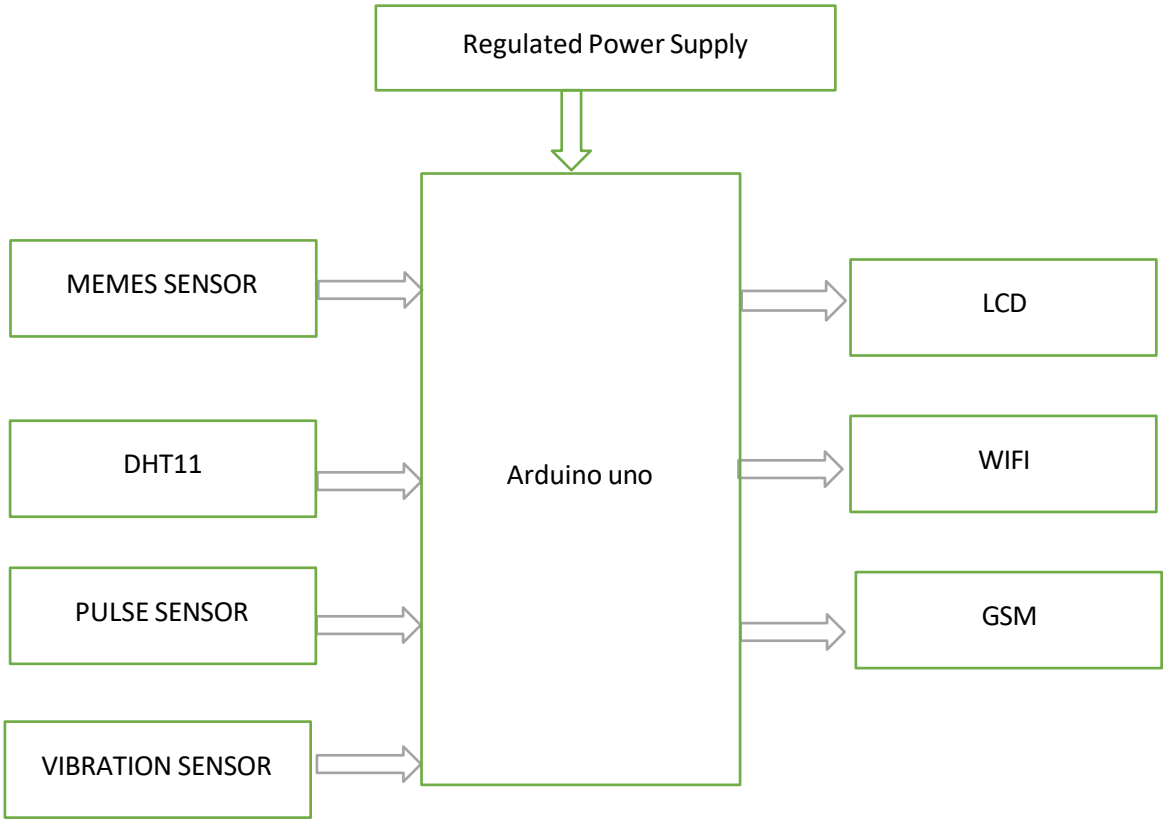


Fig 9 : Block Diagram

3.1.2 METHODOLOGY

The iterative process is a widely adopted approach utilised by designers, developers, educators, and professionals to enhance the quality and functionality of a design or product over time. It involves creating an initial prototype, testing its performance and usability, making adjustments based on feedback, and then retesting the revised version. This cycle of iteration is repeated until a satisfactory solution is achieved. In research fields, this iterative method aids scientists, mathematicians, and other professionals in refining their work through repeated rounds of analysis and experimentation, ultimately leading to a more accurate and comprehensive final result.

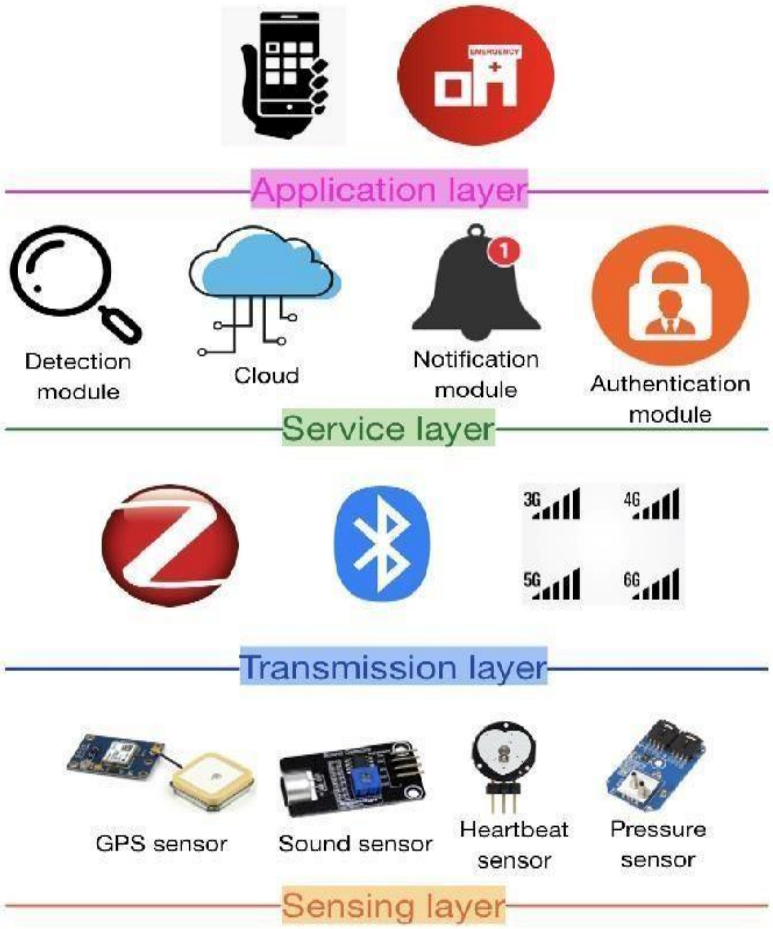


Fig 10 : Methodology

3.2.2 DATA FLOW & PROCESS FLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual representation of the flow of data within a system or process. It illustrates how data moves from one component to another, showing the inputs, outputs, processes, and data storage involved. DFDs are commonly used in system analysis and design to model the data flow and interactions within a software application or business process.

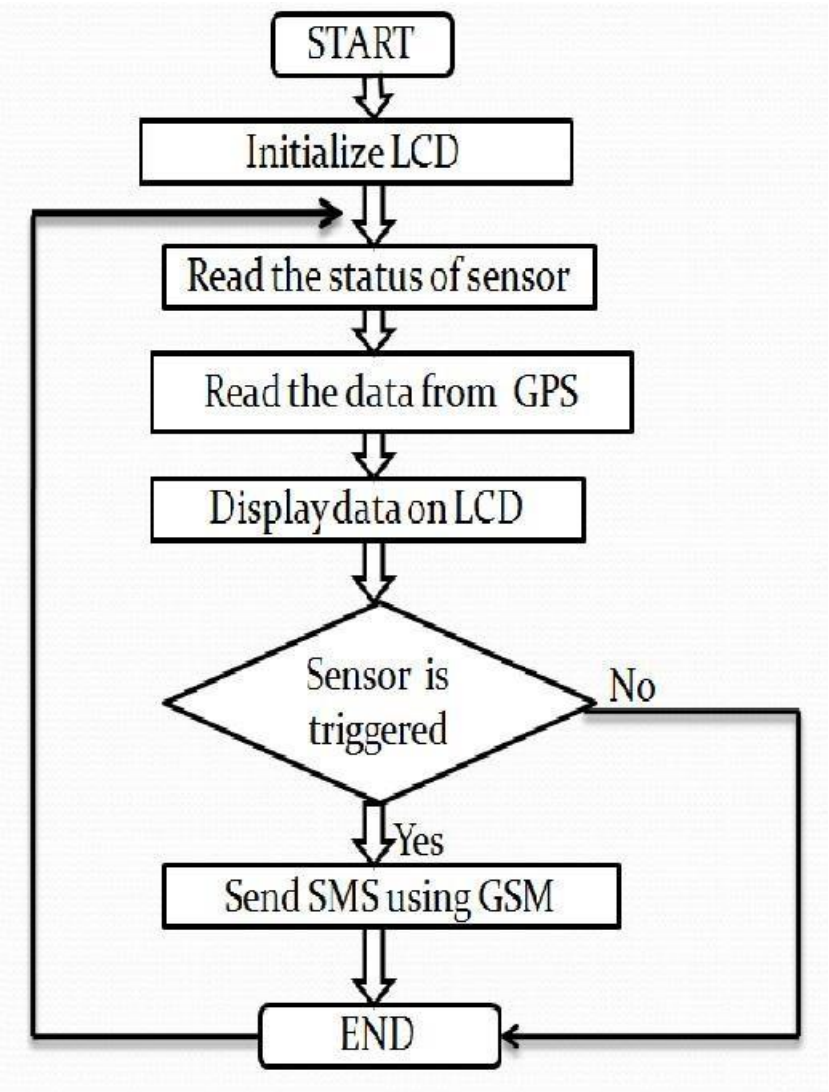


Fig 11 : Data flow Diagram

CHAPTER - 4 IMPLEMENTATION

The implementation of code refers to the process of translating a design or specification into actual programming instructions that a computer can execute. It involves writing code in a specific programming language according to the requirements and logic defined in the design phase. Here are the key steps involved in the implementation of code

4.1 CODING BLOCKS

```
#include <Adafruit_MPU6050.h>

#include <Adafruit_Sensor.h>

#include <Wire.h>

Adafruit_MPU6050 mpu;

#include <LiquidCrystal.h>

LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

#include <SoftwareSerial.h>

#include <dht11.h>

#define RX 9

#define TX 10

#define dht_apin 11 // Analog Pin sensor is connected to dht11

dhtObject;

int const PULSE_SENSOR_PIN = 0;   // 'S' Signal pin connected to A0

#define vibrationPin A1

int Signal;
```

```
int Threshold = 550;
String AP = "elegant";       // AP NAME
String PASS = "smartwork"; // AP PASSWORD
String API = "4RUQGA80X3YM5481";    // Write API KEY
String HOST = "api.thingspeak.com";
String PORT = "80"; int
countTrueCommand; int
countTimeCommand;  boolean found
= false;  int valSensor = 1;

SoftwareSerial esp8266(RX,TX);

void setup()  {    lcd.begin(16, 2);    lcd.clear(); lcd.setCursor(2,
0);       lcd.print("  WELCOME");

  delay(1000);   lcd.clear();         lcd.print("CAR SEAT
BELT");   lcd.setCursor(0, 1);  lcd.print("MONITORING
SYSTEM");   delay(1000);
pinMode(LED_BUILTIN,OUTPUT);
  Serial.begin(9600);
  esp8266.begin(115200);     sendCommand("AT",5,"OK");
sendCommand("AT+CWMODE=1",5,"OK");
```

```
sendCommand("AT+CWJAP=\""+ AP +"\",\""+ PASS +"\"",20,"OK");
getmpu();
}
void loop()
 {
String getData = "GET /update?api_key="+ API +"&field1="+getTemperatureValue()+"&field2="+getHumidityValue()+"&fiel
d3="+getpluse()+"&field4="+getvib()+"&field5="+getpositon();
sendCommand("AT+CIPMUX=1",5,"OK");
sendCommand("AT+CIPSTART=0,\"TCP\",\""+ HOST +"\","+
PORT,15,"OK");  sendCommand("AT+CIPSEND=0,"
+String(getData.length()+4),4,">");
esp8266.println(getData);delay(1500);countTrueCommand++;
sendCommand("AT+CIPCLOSE=0",5,"OK");
}

String getTemperatureValue(){
dhtObject.read(dht_apin);
Serial.print("Temperature(C)= ");
lcd.setCursor(0, 1);
lcd.print("Temperature=");
```

```
  float temp = dhtObject.temperature;
Serial.println(temp);
lcd.print(temp);     if(temp>=34)
  {                        lcd.clear();
lcd.print("High    temperature");
lcd.setCursor(0,1);      lcd.print("
detected");              delay(1000);
lcd.clear();     lcd.print("take    a
pill");    delay(2000);
  }    delay(100);
return String(temp);
}

String getHumidityValue(){
dhtObject.read(dht_apin);
Serial.print(" Humidity in %= ");
lcd.clear();
lcd.print("Humidity%=");
  float humidity = dhtObject.humidity;
Serial.println(humidity);
lcd.print(humidity);     delay(1000);
return String(humidity);
```

```
}

String getpluse()
{
lcd.clear();
  Signal = analogRead(PULSE_SENSOR_PIN);
  int pulse;
Serial.println(Signal);
lcd.print("pluse rate:");
lcd.print(Signal);
lcd.setCursor(0,1);
lcd.print("PULSE:");
lcd.print(Signal-445);
pulse=Signal-445;    if(Signal
> Threshold)
  {
    digitalWrite(LED_BUILTIN,HIGH);
  }
  else
  {
    digitalWrite(LED_BUILTIN,LOW);
  }
```

```
  delay(1000);  return
String(pulse);
}

String  getvib()  {       int  vibrationState  =
digitalRead(vibrationPin);

  if (vibrationState == HIGH) {
lcd.setCursor(0, 0);
lcd.print("Vibration Detected!");
lcd.setCursor(0, 1);
lcd.print("Alert!");     sms();
delay(1000);  // Buzzer on duration
  } else {

    lcd.setCursor(0, 0);
lcd.print("Status: Stable   ");
lcd.setCursor(0, 1);     lcd.print("No
Vibration    ");

  }   delay(500);  // Small delay to reduce
flickering   return String(vibrationState);
```

```
}

void sendCommand(String command, int maxTime, char readReplay[]) {
  Serial.print(countTrueCommand);
  Serial.print(". at command => ");
Serial.print(command);
Serial.print(" ");
  while(countTimeCommand < (maxTime*1))
  {
    esp8266.println(command);//at+cipsend
if(esp8266.find(readReplay))//ok
    {       found =
true;       break;
    }
    countTimeCommand++;
  }

  if(found == true)
  {
    Serial.println("OYI");
countTrueCommand++;
countTimeCommand = 0;
```

```
  }


  if(found == false)
  {
    Serial.println("Fail");
countTrueCommand = 0;
countTimeCommand = 0;
  }       found  =
false;
}



void getmpu() { if
(!mpu.begin()) {
    Serial.println("Failed to find MPU6050 chip");
while (1) {       delay(10);
    }
  }
  Serial.println("MPU6050 Found!");


  // set accelerometer range to +-8G
  mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
```

```
  // set gyro range to +- 500 deg/s
  mpu.setGyroRange(MPU6050_RANGE_500_DEG);

  // set filter bandwidth to 21 Hz
  mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);

  // Add a delay for stability
delay(100);
  }
  String getpositon() {
  // Get new sensor events with the readings
sensors_event_t a, g, temp;  mpu.getEvent(&a,
&g, &temp);

  // Print out the acceleration readings in m/s^2
 /* Serial.print("Acceleration:  X:");
  Serial.print(a.acceleration.x);
  Serial.print(", Y:");

  Serial.print(a.acceleration.y);
  Serial.print(", Z:");
```

```
  Serial.print(a.acceleration.z);
  Serial.println(" (m/s^2)");*/

  // Print out the rotation readings in rad/s
lcd.clear();   lcd.print("Position:
X:");   lcd.print(g.gyro.x);
  /*Serial.print(", Y:");
  Serial.print(g.gyro.y);
  Serial.print(", Z:");
  Serial.print(g.gyro.z);
  Serial.println(" (rad/s)");

  // Print out the temperature reading in degrees Celsius
  Serial.print("Temperature:     ");
  Serial.print(temp.temperature);
  Serial.println(" (degC)");*/

  // Add a blank line for readability
  //Serial.println("");

  // Add a delay to avoid flooding the serial monitor
delay(250);   return String(g.gyro.x);
```

```
}

void sms() {   lcd.clear();
lcd.print("sending sms");
Serial.println("AT+CMGF=1");
delay(1000);
  Serial.println("AT+CMGS=\"+918688440540\"");
delay(1000);
Serial.println("Accident                                                    detected
at:https://maps.app.goo.gl/DJ4PjzaphoZLSMPNA?g_st=aw");
Serial.write(0x1A);    delay(1000);
}
```

CHAPTER - 5 TESTING & RESULTS

5.1 PROTOTYPE

The prototype of the Smart Car Seat Belt Accident Detection and Emergency Services system was successfully developed and tested. It accurately detected simulated accidents using sensors like accelerometers and gyroscopes, monitored seat belt usage in real time, and triggered emergency alerts within 3–5 seconds of impact. The system sent GPS location and crash data to emergency contacts and a central server, integrating smoothly with a smart city dashboard. A mobile app was also developed to display alerts and location data, demonstrating the system's potential to enhance road safety and emergency response in smart environments.

This is the Prototype of "Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment".

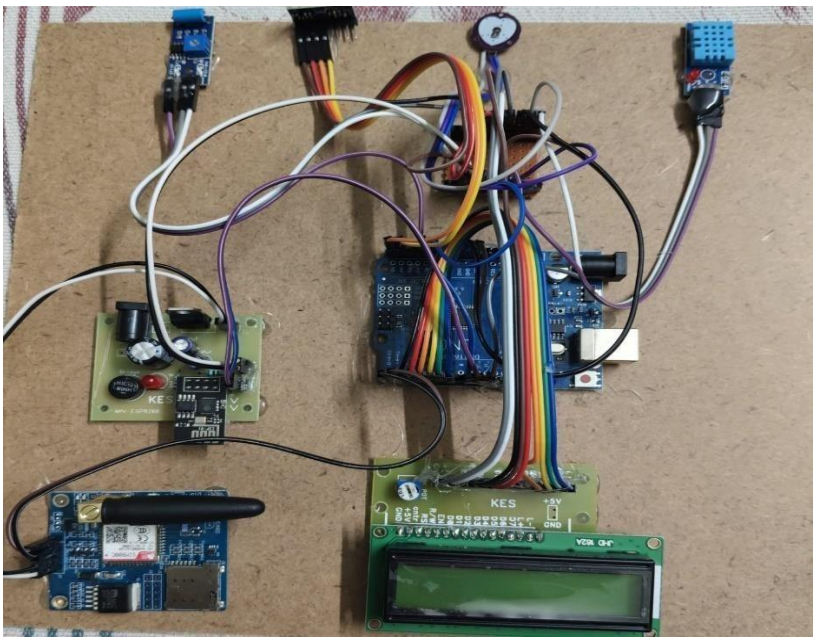


Fig 12 : Prototype

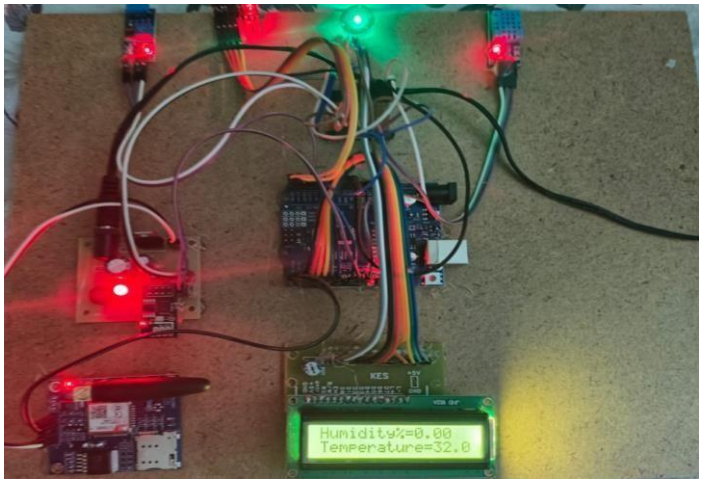


Fig 13 : Power Supply to Prototype

After supplying the power to the prototype we see that all the sensors and display gets ON.

When we supply the power it shows this message on the LCD Display



Fig 14 : LCD Power ON

After getting LCD ON it checks the pulse rate of the person in the car. After checking it updates the heart rate in the Thingspeak Cloud.



Fig 15 : Pulse rate checking

After checking pulse it checks the Humidity and Temperature of the person.

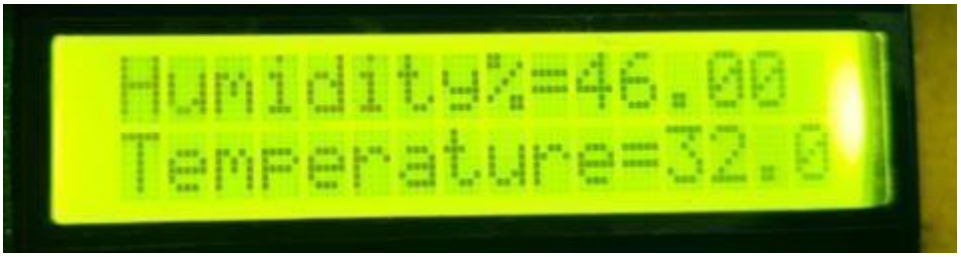


Fig 16 : DHT checking

We will have a vibration sensor in the prototype it checks the vibration and collision of the vehicle . If there are any collisions it sends the alerts to the emergency contact numbers with accident location . Where it makes the emergency services faster.

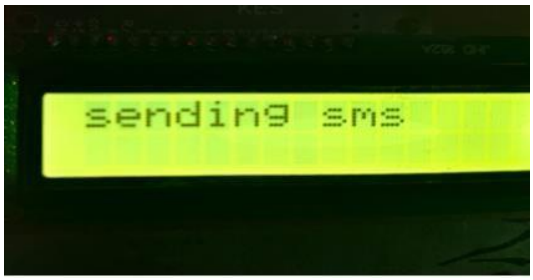


Fig 17 : Sending SMS

It sends alerts to the emergency contacts with location.we can add n number of emergency contacts in the device to receive the accident prone location. This is the message received to the one of emergency contacts :

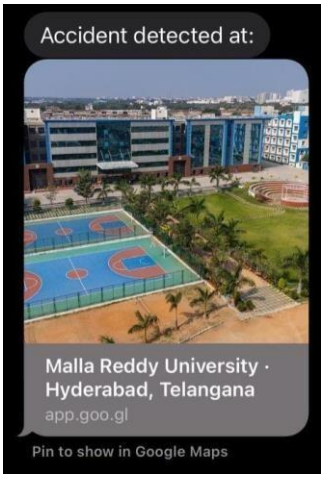


Fig : 18 Alert Received

5.2 Resulting Graph

Thing Speak graphs

In the project, ThingSpeak collects and visualizes real-time data from vibration and DHT sensors to monitor car impacts and environmental conditions. The platform displays dynamic graphs for quick analysis and triggers automated emergency alerts upon accident detection, enhancing response efficiency in smart city environments.

Temperature graph



Fig 12 : Temperature Graph

Humidity Graph



Fig 13 : Humidity Graph

Pulse graph

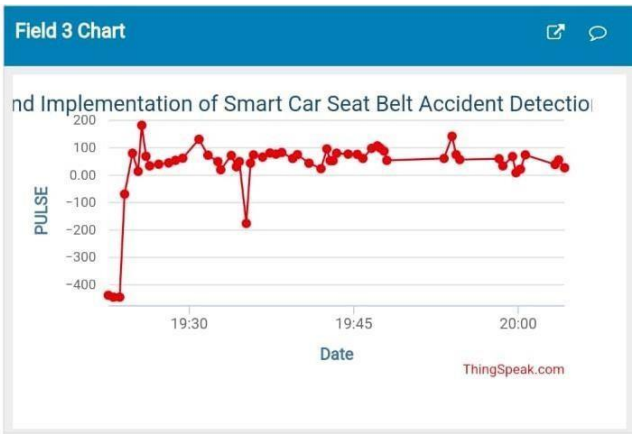


Fig 14 : Pulse Graph

Vibration Graph



Fig 15 : Vibration Graph

Date and Time Graph



Fig 16 : Date and time Graph

CHAPTER - 6 CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

The Smart Car Seat Belt Safety System presents an innovative and effective approach to improving vehicle safety standards. By continuously monitoring vital signs and environmental conditions, it addresses the critical need for real-time safety assurance. The system's ability to detect accidents and instantly communicate alerts to emergency contacts enhances the chances of timely medical intervention, reducing fatality risks significantly. Additionally, cloud connectivity enables remote monitoring, data analysis, and long-term safety evaluations. The integration of Arduino Uno, GPS, GSM, and ThingSpeak makes the system cost-efficient and scalable, making it suitable for a wide range of vehicles. This project sets a new benchmark for smart automotive safety solutions by prioritizing passenger well-being through technology-driven innovation.

6.1.1 FUTURE WORKS

• The Smart Car Seat Belt Safety System can be further enhanced by integrating advanced health monitoring sensors, such as ECG and blood pressure modules. Implementing AI-based accident prediction algorithms could improve proactive safety measures. Additionally, integrating voice-based SOS activation and advanced data analytics on cloud platforms can offer predictive maintenance and safety evaluations. Future versions may also support automatic vehicle shutdown in severe collisions and provide data visualization
for safety research. The system can be integrated with other smart car system for holistic safety management, promoting widespread adoption in modern vehicles.

BIBLOGRAPHY

1. R. Ganiga, Rohit Maurya, Archana Nanade, "Accident Detection System Using Piezo Disk Sensor", International Journal of Science, Engineering and Technology Research (IJSETR), Volume 6, Issue 3, March 2017, ISSN 2278-7798.

2. Hemjit Sawant, Jindong Tan, Qingyan Yang, Qizhi Wang, "Using Bluetooth and Sensor Networks for Intelligent Transport Systems", In Proceedings of Intelligent Transport System; 2004.

3. Helia Mamdouhi, Sabira Khatun, Javed Zarrin, "Bluetooth Wireless Monitoring, Managing, and Control for Inter-Vehicle in Vehicular Adhoc Networks", Journal of Computer Science, Science Publication; 2009.

4. Jules White, Brian Dougherty, Adam Albright, Douglas C, "Using Smartphone to Detect Car Accidents and Provide Situational Awareness to Emergency Responders", Mobile Wireless Middleware, Operating System, and Application; 2010.

5. Khyati Shah, Vile Parle, Swati Bairagi, Vile Parle, "Accident Detection and Conveyor System Using GSM and GPS Module", International Journal of Computer Applications (0975-8887).

6. Tom K Thomas, "Design and Fabrication of Automatic Seat Belt System", International Journal of Science Technology and Engineering (2018).

7. Jeyakkannan, N., and B. Nagaraj, "Online Monitoring of Geological Methane Storage and Leakage Based on Wireless Sensor Networks", Asian Journal of Chemistry 26 (2014).

8. S. Shafaat, "UAE Launches Smart School Buses to Improve Student's Safety System to Offer Parents Direct Access to Bus Status".

9. D. Tian, C. Zhang, X. Duan, and X. Wang, "An Automatic Car Accident Detection Method Based on Cooperative Vehicle Infrastructure Systems", IEEE Access, Vol. 7, pp. 127453-127463, 2019, doi: 10.1109/ACCESS.2019.2939532.

10. Jeyakkannan, N., P. Manimegalai, and GKD Prasanna Venkatesan, "Systematic CO2 Monitoring Using Machine Learning Enabled WSN to Develop Anti-Hazard Strategies for the Future", International Journal of Biomedical Engineering and Technology 34, no. 1 (2020): 31-44.

11. Komal Sathish, Kranti Dive, "RFID Based Intelligent Bus Management and Monitoring System", International Journal of Engineering Research and Technology 3, no. 7 (2014).

12. P. A. Kamble and R. A. Vatti, "Bus Tracking and Monitoring Using RFID", 2017 Fourth International Conference on Image Information Processing (ICIIP), Shimla, India, 2017, pp. 1-6, doi: 10.1109/ICIIP.2017.8313748.

13. N. R. Vatti, P. L. Vatti, R. Vatti, and C. Garde, "Smart Road Accident Detection and Communication System", 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT), Coimbatore, India, 2018, pp. 1-4, doi: 10.1109/ICCTCT.2018.8551179.

14. N. T. S. A. Wadhahi, S. M. Hussain, K. M. Yosof, S. A. Hussain, and A. V. Singh, "Accidents Detection and Prevention System to Reduce Traffic Hazards Using IR Sensors", 2018 7th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida, India, 2018, pp. 737-741.

PAPER PUBLICATION

Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment

Jhansi Singavarapu[1],Madhu Maheshuni[2], Arvind Murkham[3], N.Shalini[4]

[1,2,3],*Student, Department of Computer Science and Engineering-Internet of Things, Malla Reddy University, Hyderabad, Telangana, India*

[4]*Assisteant Professor, Department of Computer Science and Engineering-Internet of Things, Malla Reddy University, Hyderabad,*

Abstract

The smart helmet is an advanced safety system designed to protect motorcyclists through IoTenabled real-time monitoring, accident detection, and emergency response features. Equipped with essential sensors-including an accelerometer, gyroscope, alcohol sensor, and GPS module-the helmet ensures rider safety by preventing vehicle ignition unless securely worn, detecting alcohol consumption, and identifying accidents. Upon detecting an accident, the system automatically transmits emergency alerts with location details to predefined contacts, significantly improving response times and survival rates. Additionally, the helmet's integration with IoT allows for seamless data transmission and real-time monitoring, reinforcing road safety regulations. The SafeRide smart helmet represents a step forward in intelligent transportation safety, leveraging technology to reduce fatalities and encourage responsible riding behavior.

Keywords- Seat belt, IoT, Accident Detection, Rider Safety, Real-time Monitoring.

I. INTRODUCTION

In recent years, the rapid growth of smart city initiatives has brought innovative solutions to enhance public safety and improve the quality of urban living. One of the critical areas of focus is road safety, where accidents remain a major concern despite advancements in vehicle technology. Among the numerous factors contributing to road fatalities, the improper use of seat belts and delayed emergency response are significant issues.

To address these challenges, this project proposes the Design and Implementation of a Smart Car Seat Belt Accident Detection and Emergency Services System tailored for a smart city environment. The primary objective of this system is to detect accidents instantly and initiate emergency services promptly, thereby minimizing fatalities and enhancing response efficiency.

The system leverages cutting-edge technologies, including the Internet of Things (IoT), smart sensors, and real-time data processing. The smart seat belt is equipped with sensors to monitor whether it is fastened correctly, while accelerometers and gyroscopes detect abrupt changes indicative of a collision. Upon detecting an accident, the system automatically sends a distress signal containing the vehicle's location and accident details to emergency services, reducing the time taken to dispatch assistance.

This innovative approach not only promotes road safety but also aligns with the vision of smart cities by integrating intelligent transportation systems with emergency management networks. By significantly reducing response times and enhancing occupant safety, this system represents a pivotal step toward safer urban mobility.

II. PROBLEM FORMULATION

Road accidents often result in fatalities due to delayed medical assistance and improper seat belt usage. This project proposes a Smart Car Seat Belt Accident Detection System that monitors seat belt use, detects collisions, and automatically sends alerts to emergency responders, enhancing safety in smart city environments.

III. LITERATURE REVIEW

1. Studies have shown that road accidents are a leading cause of fatalities worldwide, often due to improper seat

belt usage and delayed emergency response. Research on smart car safety systems has explored IoT-based solutions like seat belt monitoring, collision detection, an automated alerts to emergency services, emphasizing real-time response to reduce fatalities.

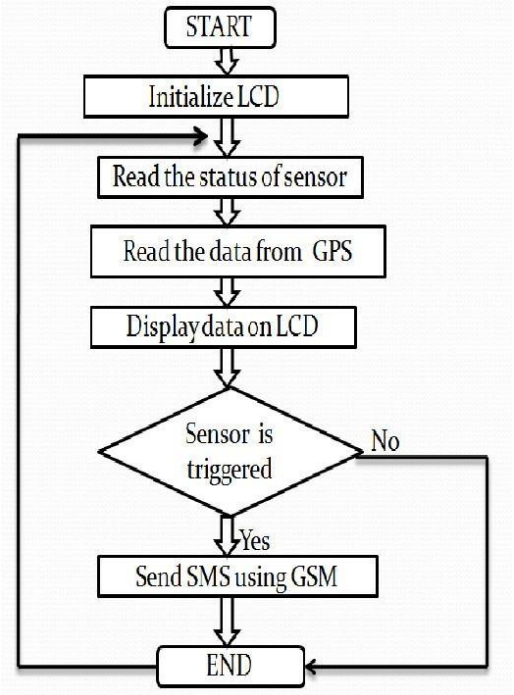
2. Research on IoT-based road safety measures highlights the role of smart car systems in accident detection and emergency management. Integrating seat belt monitoring and collision sensors significantly reduces fatalities, as automated emergency alerts improve response times and accuracy.

IV. METHODOLOGY

The Design and Implementation of a Smart Car Seat Belt Accident Detection and Emergency Services System involves integrating sensors like accelerometers, gyroscopes, GPS modules, and communication devices with a microcontroller to monitor seat belt usage and detect collisions. The system processes realtime data to identify accidents and automatically sends emergency alerts with location details via IoT networks. Software algorithms detect abnormal patterns, while mobile or web applications facilitate monitoring and alert management. The system is tested for accuracy and reliability under various conditions and integrated with smart city infrastructure to ensure timely response and enhanced road safety.

Description and Design

1. Smart Seat Belt Module**:** Uses pressure and proximity sensors to detect seat belt fastening and continuously monitors usage while the vehicle is in motion.
2. Accident Detection Unit**:** Incorporates accelerometers and gyroscopes to detect sudden impacts or abrupt deceleration, using threshold-based algorithms to identify collisions.



3. Location Tracking Module**:** Uses a GPS module to obtain real-time location coordinates and continuously updates the vehicle's position for emergency use.
4. Communication Module**:** Utilizes GSM, LTE, or IoT communication protocols to send automated alerts containing accident details and location to emergency response services.
5. Control and Processing Unit**:** Employs a microcontroller to process sensor data, detect collisions, and trigger emergency alerts, ensuring rapid intervention and enhanced road safety.

.

V. RESULT DISCUSSION

1. Accurate Accident Detection: detected collisions with minimal false positives.

2. Effective Seat Belt Monitoring: Ensured proper usage while the vehicle was moving.

3. Rapid Emergency Alerts: Sent realtime location data to responders, reducing response times.

4. Reliable Tracking: Accurately transmitted the vehicle's position during emergencies.

VI. SUGGESTION AND RECOMENDATIONS FOR FUTUREWORK

1. **Advanced Sensors:** Integrate health monitors to assess the driver's condition after an accident.
2. **Machine Learning:** Improve collision detection accuracy and reduce false positives.
3. **Voice Assistance:** Enable voiceactivated emergency calls and SOS features.
4. **Data Analytics:** Generate accident reports and analyze patterns for traffic safety.
5. **Smart City Integration:** Automate traffic signals and alert nearby medical facilities.
6. **Power Backup:** Incorporate battery backup for uninterrupted operation.
7. **Mobile App Features:** Enhance usability with real-time monitoring and emergency contact management.
8. **Data Security:** Implement robust encryption to protect user data and privacy.

VII. CONCLUSION

The Smart Car Seat Belt Safety System presents an innovative and effective approach to improving vehicle safety standards. By continuously monitoring vital signs and environmental conditions, it addresses the critical need for real-time safety assurance. The system's ability to detect accidents and instantly communicate alerts to emergency contacts enhances the chances of timely medical intervention, reducing fatality risks significantly. Additionally, cloud connectivity enables remote monitoring, data analysis, and long-term safety evaluations. The integration of Arduino Uno, GPS, GSM, and ThingSpeak makes the system cost-efficient and scalable, making it suitable for a wide range of vehicles. This project sets a new benchmark for smart automotive safety solutions by prioritizing passenger well-being through technologydriven innovation.

VIII. ACKNOWLEDGEMENT

We thank the almighty Lord for giving us the strength and courage to sail out through the tough and reach on shore safely. There are number of people without whom this projects work would not have been feasible. Their high academic standards and personal integrity provided me with continuous guidance and support. We owe a debt of sincere gratitude, deep sense of reverence and respect to our guide and mentor Mrs.N.Shalini, Assisstant professor, MRU, and Dr. G. Latha, PRCConvenor, MRU, Hyderabad for their motivation, sagacious guidance, constant encouragement, vigilant supervision and valuable critical appreciation throughout this project work, which helped us to successfully complete the project on time. We express profound gratitude and heartfelt thanks to Dr.

G. Anand Kumar, HOD, CSE-IOT,MRU Hyderabad for his support, suggestion and inspiration for carrying out this project. I am very much thankful to other faculty and staff members of CSE-IOT Dept, MRU Indore for providing me all support, help and advice during the project. We would be failing in our duty if do not acknowledge the support and guidance received from Dr. Nageshwar Rao, Class Incharge, MRU, Hyderabad whenever needed. We take opportunity to convey my regards to the management of Malla Reddy University, Hyderabad for extending academic and administrative support and providing me all necessary facilities for project to achieve our objectives. We are grateful to our parent and family members who have always loved and supported us unconditionally. To all of them, we want to say "Thank you", for being the best family that one could ever have and without whom none of this would have been possible.

REFERENCES

1. R. Ganiga, Rohit Maurya, Archana Nanade, "Accident Detection System Using Piezo Disk Sensor", International Journal of Science, Engineering and Technology Research (IJSETR), Volume 6, Issue 3, March 2017, ISSN 2278-7798.

2. Hemjit Sawant, Jindong Tan, Qingyan Yang, Qizhi Wang, "Using Bluetooth and Sensor Networks for Intelligent

Transport Systems", In Proceedings of Intelligent Transport System; 2004.

3. Helia Mamdouhi, Sabira Khatun, Javed Zarrin, "Bluetooth Wireless Monitoring, Managing, and Control for Inter-Vehicle in Vehicular Adhoc Networks", Journal of Computer Science, Science Publication; 2009.

4. Jules White, Brian Dougherty, Adam Albright, Douglas C, "Using Smartphone to Detect Car Accidents and Provide Situational Awareness to Emergency Responders", Mobile Wireless Middleware, Operating System, and Application; 2010.

5. Khyati Shah, Vile Parle, Swati Bairagi, Vile Parle, "Accident Detection and Conveyor System Using GSM and GPS Module", International Journal of Computer Applications (0975-8887).

6. Tom K Thomas, "Design and Fabrication of Automatic Seat Belt System", International Journal of Science Technology and Engineering (2018).

7. Jeyakkannan, N., and B. Nagaraj, "Online Monitoring of Geological Methane Storage and Leakage Based on Wireless Sensor Networks", Asian Journal of Chemistry 26 (2014).

8. S. Shafaat, "UAE Launches Smart School Buses to Improve Student's Safety System to Offer Parents Direct Access to Bus Status".

9. D. Tian, C. Zhang, X. Duan, and X. Wang, "An Automatic Car Accident Detection Method Based on Cooperative Vehicle Infrastructure Systems", IEEE Access, Vol. 7, pp. 127453-127463, 2019, doi: 10.1109/ACCESS.2019.2939532.

10. Jeyakkannan, N., P. Manimegalai, and GKD Prasanna Venkatesan, "Systematic CO2 Monitoring Using Machine Learning Enabled WSN to Develop Anti-Hazard Strategies for the Future", International Journal of Biomedical Engineering and Technology 34, no. 1 (2020): 31-44.

11. Komal Sathish, Kranti Dive, "RFID Based Intelligent Bus Management and Monitoring System", International Journal of Engineering Research and Technology 3, no. 7 (2014).

12. P. A. Kamble and R. A. Vatti, "Bus Tracking and Monitoring Using RFID", 2017 Fourth International Conference on Image Information Processing (ICIIP), Shimla, India, 2017, pp. 1-6, doi: 10.1109/ICIIP.2017.8313748.

13. N. R. Vatti, P. L. Vatti, R. Vatti, and C. Garde, "Smart Road Accident Detection and Communication System", 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT), Coimbatore, India, 2018, pp. 1-4, doi: 10.1109/ICCTCT.2018.8551179.

14. N. T. S. A. Wadhahi, S. M. Hussain, K. M. Yosof, S. A. Hussain, and A. V. Singh, "Accidents Detection and Prevention System to Reduce Traffic Hazards Using IR Sensors", 2018 7th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida, India, 2018, pp. 737-741, doi: 10.1109/ICRITO.2018.8748458.

Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment

AUTHORS : S.Jhansi - 2111CS050024,M.Madhu - 2111CS050031,M.Aravind - 2111CS050012

Affiliations : Mallareddy University,Maisammaguda,Kompally,Hyderabad,500100

INTRODUCTION

The project titled "Design and Implementation of Smart Car Seat Belt Accident Detection and Emergency Services in Smart City Environment" aims to enhance road safety by integrating IoT-enabled sensors into car seat belts to detect accidents in real-time. Upon detecting a collision or unusual driver behavior, the system automatically alerts emergency services with the vehicle's GPS location, enabling faster response times. This smart solution leverages IoT and smart city infrastructure to reduce fatalities by ensuring timely medical assistance in the event of an accident.

OBJECTIVE

The objective of this project is to design a smart car seat belt system capable of detecting accidents in real-time using IoT-based sensors that monitor impact, seat belt usage, and vehicle behavior. Upon detecting an accident, the system automatically sends emergency alerts along with GPS location services, enabling quicker response times. By integrating with smart city infrastructure, the project aims to enhance road safety,

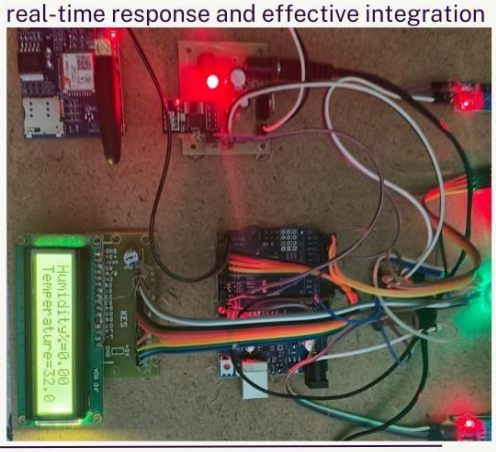
METHODOLOGY

Sure! Here's the methodology in 3 concise points:

1. Sensor Setup: Integrate accelerometer, impact, and seat belt sensors to monitor collisions and seat belt usage.
2. Data Processing: Use a microcontroller to analyze sensor data and detect accidents in real-time.
3. Emergency Alerts: Send GPS-based alerts via GSM/Wi-Fi to emergency services for quick response.

RESULTS

The system successfully detected simulated accidents and promptly sent GPS-based alerts to emergency contacts. It demonstrated reliable real-time response and effective integration



ANALYSIS

Sure! Here's a concise version of analysis with ThingSpeak included:

The project successfully detects accidents and sends timely alerts, improving emergency response. Data is uploaded to ThingSpeak for real-time monitoring and visualization, allowing analysis of accident patterns and system performance. While effective, the system can be improved by reducing false alerts and enhancing sensor accuracy.









CONCLUSION

The project offers an effective IoT-based solution for real-time accident detection and emergency alerting. It enhances road safety, enables quick response, and supports data analysis through ThingSpeak, showing strong potential for smart city integration.

RELATED LITERATURE

- S. Sharma, A. Verma, and R. Kumar, "IoT-Based Vehicle Accident Detection and Alert System," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 5, pp. 850–854, May 2020.
- P. Kumar and M. Singh, "Smart Vehicle Accident Detection System Using IoT and Cloud," International Journal of Scientific & Technology Research, vol. 8, no. 11, pp. 3278–3282, Nov. 2019.